MILITARY SPECIFICATION

RADIATION SOURCE, INFRARED, 2.2 KW FOR SEARCHLIGHT, INFRARED, 28-VOLT, 110-AMPERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of short-arc, xenon radiation source, infrared.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-76	Tape, Pressure Sensitive-Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure Sensitive-Adhesive Filament Reinforced

MILITARY

MIL-P-116	Preservation Packaging, Methods
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-P-11268	Parts, Materials, and Processes Used in Electronics Equipment
MIL-M-13231	Marking of Electronic Items
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-613912	Radiation Source, Infrared and Container Assembly

(Copies of specifications, standards, and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN WELDING SOCIETY

Standard Qualification Procedure

Application for copies should be addressed to:

American Welding Society
435 East 47th Street
New York, New York 10016

3. REQUIREMENTS

3.1 Description. The radiation source, infrared, hereinafter known as the "source", shall be a gaseousdischarge, shortarc source utilizing xenon gas in a quartz envelope and fitted with a locating and supporting framework and electrical connectors.

3.2 Construction. The source shall be constructed in accordance with the requirements of this specification and Drawing SC-DL-613912.

3.3 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

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3.5 Operational and environmental.

3.5.1 Starting and operating. The source shall ignite and maintain ignition within 3 seconds, as specified herein. The operating power shall be supplied on a continuous basis with the ignition and sustaining power applied, nonrecurring, in any one ignition attempt, until boost current exceeds 30 amperes. The ignition voltage required to cause an arc between the electrodes shall be not less than 18 Kilovolt (kV) sustained. Using booster characteristics curves as shown in Figures 1 and 2, the sustaining voltage shall be 70 to 110 volts dc of an exponential decay function. The maintaining voltage shall be not less than 16 volts dc, nor more than 22 volts dc.

3.5.2 Normal operation. The source shall ignite and operate as specified herein with a voltage of 22.0 volts, plus or minus 0.5 volts dc, at a current of 100 amperes.

3.5.3 Overdrive operation. The source shall operate for one 20 second period in each 5 minutes with a voltage of 25 to 26 volts dc, and a source current of 125 to 150 amperes.

3.5.4 Brightness. The average brightness of the arc over an area 4.5 millimeters vertically by 2.5 millimeters horizontally shall be not less than 400 candles per square millimeter with the source operating at 100 amperes.

3.5.5 Lumen output. Initial lumen output of the source shall be not less than 58,000 lumens when the source is operated at 100 amperes.

3.5.6 Infrared output. The infrared output of the source shall be not less than 10 percent of the total light output.

3.5.7 Mechanical shock. The source shall not be damaged (see 6.3) by mechanical shocks (using a half sine shock pulse configuration Method 516 of MIL-STD-810) of 15 g's (see 6.3) with each shock having a duration of 6 milliseconds, plus or minus 1.0 millisecond.

3.5.8 Vibration. The source shall not be damaged by vibration throughout a frequency range of 15 to 120 hertz (Hz) at a value of 2.5 g's, plus or minus 0.5 g.

3.5.9 Altitude. The source shall ignite and operate as specified herein at 8,000 feet above sea level with ambient temperature maintained at plus 35°C.

3.5.10 Altitude transportation. The source shall not be damaged by transportation in unpressurized aircraft at any altitude to 40,000 feet above sea level at corresponding temperature for this altitude.

3.5.11 Storage temperature.

3.5.11.1 High temperature storage. The source shall not be damaged by storage at any ambient temperature from plus 23°C to plus 68°C with humidity controlled at 90 to 98 percent for 6 hours.

3.5.11.2 Low temperature storage. The source shall not be damaged by storage at any ambient temperature from plus 23°C to minus 62°C for 6 hours.

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3.5.12 Operating temperatures.

3.5.12.1 High temperature operation. The source shall operate as specified herein at any ambient temperature to plus 52°C with humidity uncontrolled.

3.5.12.2 Low temperature operation. The source shall operate as specified herein at any ambient temperature to minus 54°C with humidity uncontrolled.

3.5.13 Quartz bulb. The optical quality of the quartz bulb shall be such that the apparent shift of the cathode observed in any direction shall be within a circle of 1.0 millimeter radius. The quartz bulb shall be free from imperfections such as bubbles, scratches, and visible distortions which may impair the operation of the source or safety of personnel.

3.5.14 Electromagnetic compatibility. The electromagnetic compatibility of the source shall conform to MIL-STD-461, Table A-1-MIL-E-55301, equipment class tactical non-CE.

3.5.15 Life. The source shall have a lumen output of not less than 70 percent of its initial lumen output at the end of 300 hours of operation.

3.6 Identification marking. The source shall be marked in accordance with MIL-M-13231.

3.7 Workmanship. All parts, components and assemblies of the source including forgings, molded parts, stampings, seals, welded parts and machined surfaces shall be clean and free from sand, dirt, pits, sprues, scale, flux and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges and corners are required.

3.7.1 Welding. Welding shall conform to standard welding practice as prescribed by the American Welding Society, (see 2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified by the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirement as specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and materials inspection. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as specified herein.

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4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

c. Inspection of preparation for delivery (see 4.7).

4.3 First article inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 Inspection. Each source shall be examined in accordance with Table I. One or more defects shall be cause for rejection of that source.

TABLE I Inspection

Defects		Requirement paragraph
Major		
101.	Dimensions not as specified	3.2
102.	Parts and Materials not as specified	3.4
103.	Container not furnished or not as specified	3.4
104.	Marking not in accordance with MIL-M-13231	3.6
105.	Workmanship not as specified	3.7
Minor		
	Defects not listed as major shall be classified as minor.	

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1, the first article source shall be subjected to the inspections in Table II. Inspections may be conducted in any sequence. Failure of any test shall be cause for rejection.

TABLE II First article inspection

Inspection	Requirement paragraph	Test paragraph
Starting and operating	3.5.1	4.6.1
Normal operation	3.5.2	4.6.2
Overdrive operation	3.5.3	4.6.3
Brightness	3.5.4	4.6.4
Lumen output	3.5.5	4.6.5
Infrared output	3.5.6	4.6.6
Mechanical shock	3.5.7	4.6.7
Vibration	3.5.8	4.6.8

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TABLE II First article inspection (con't)

Inspection	Requirement paragraph	Test paragraph
Altitude	3.5.9	4.6.9
Altitude transportation	3.5.10	4.6.10
High temperature storage	3.5.11.1	4.6.11.1
Low temperature storage	3.5.11.2	4.6.11.2
High temperature operation	3.5.12.1	4.6.12.1
Low temperature operation	3.5.12.2	4.6.12.2
Quartz bulb	3.5.13	4.6.13
Electromagnetic compatibility	3.5.14	4.6.14
Life	3.5.15	4.6.15

4.4 Inspection covered by subsidiary documents: The source shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order (see 4.3.1, Table I).

4.5 Quality conformance inspection.

4.5.1 Inspection. Each source shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests.

4.5.2.1 Group A Inspection. Each source shall be inspected in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Starting and operating	3.5.1	4.6.1
Brightness	3.5.4	4.6.4

4.5.2.2 Group B Inspection. Group B inspection shall be conducted on sources selected from units which have passed the examination and tests in 4.5.1 and 4.5.2.1. Sampling shall be in accordance with MIL-STD-105, AQL 6.5 percent defective, inspection level S-2. Tests shall be conducted in accordance with Table IV, and may be performed in any sequence.

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TABLE IV Group B Inspection

Inspection	Requirement paragraph	Test paragraph
Normal operation	3.5.2	4.6.2
Lumen output	3.5.5	4.6.5
Infrared output	3.5.6	4.6.6
Quartz bulb	3.5.13	4.6.13

4.5.2.3 Group C Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.2.3.1.

4.5.2.3.1 Sampling for inspection. One sample source shall be selected at random by the Government for all inspections in Table V from the separate monthly specified quantity (see 6.2) produced or each 100 sources, whichever occurs first. The sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE V Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Mechanical shock	3.5.7	4.6.7
Vibration	3.5.8	4.6.8
Low temperature storage	3.5.11.2	4.6.11.2
High temperature operation	3.5.12.1	4.6.12.1
Low temperature operation	3.5.12.2	4.6.12.2
Life	3.5.15	4.6.15

4.5.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order, (see 6.2).

4.5.4 Reinspection of Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C reinspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test methods. Unless otherwise specified in the individual tests, airflow and air temperature for all tests shall be 40 to 45 cubic feet per minute. Forced-air cooling shall be directed on the top of the source in a downward direction parallel to the electrode axis. The source shall be considered as stabilized for all light measurements after 5 minutes operation with cooling airflow. Unless otherwise specified in the individual tests, the source shall be operated at 100 amperes. All meters used during test shall

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have been calibrated within 30 days of the start of testing, and thereafter as required, and at intervals not to exceed 6 months. Test shall be monitored by direct-reading, laboratory-type meters accurate to within 0.5 percent and connected to indicate in the most accurate portion of the range. Voltage, current drain, ambient, and operating temperature during environmental testing shall be recorded using recording instruments, accurate within 3.0 percent with a damping factor not greater than 3 as defined in MIL-M-10304. Test facilities, chambers and apparatus used in conducting tests specified shall meet the requirements of MIL-STD-810. All voltage measurements shall be made only across radiation source electrodes.

4.6.1 Starting and operation. The source shall be ignited with an operating voltage of 16 and 22 volts dc. The source shall maintain ignition for 5 minutes in each of the above voltage levels. The operating voltage shall be supplied on a continuous basis with the ignition and sustaining power applied, nonrecurring in any one ignition attempt, until boost current exceeds 30 amperes. Failure of the source to ignite and maintain ignition within 3 seconds, or failure of the source to conform to the booster characteristics curves as shown in Figures 1 and 2 shall constitute failure of this test.

4.6.2 Normal operation. The source shall be operated for 20 minutes. At the end of the 20 minute operating period the voltage and current shall be recorded. Failure of the source to meet the specified voltage and current in 3.5.2 shall constitute failure of this test.

4.6.3 Overdrive operation. The source shall be operated for 20 minutes. Once every 5 minutes during the 20 minute operating period the source voltage shall be raised to 25 to 26 volts dc. Failure of the source to achieve 125 to 150 amperes when the voltage is at 25 to 26 volts dc shall constitute failure of this test.

4.6.4 Brightness. The source shall be supported in the vertical position and operated. The average brightness of the arc shall be measured over an area 4.5 millimeters vertically and 2.5 millimeters horizontally using a lens with a stop of known areas and a photocell corrected to a photopic-eye-response curve. The photocell shall be calibrated with a tungsten lamp of 2,000 watts operated at a color temperature of 2870° Kelvin (K), plus or minus 50° K. An average brightness less than 400 candles per square millimeter shall constitute failure of this test.

4.6.5 Lumen output. The source shall be supported in an integrating sphere having a minimum diameter of 3 feet and shall be operated in open air. Total lumens shall be measured during the first 10 hours of operation, using standard photometric techniques as follows: Precautions shall be taken to prevent supports and baffles from absorbing light or affecting the source temperature. There shall be no selective spectral absorption by the sphere, sphere window, or internal support. The source shall be positioned in the center of the sphere and forced-air cooling shall be used to limit the temperature of the source to its operating range. The forced-air cooling shall be directed on the top of the source downward direction parallel to the electrode axis. The light-measuring device shall receive no direct light from the source and shall be color-corrected for photopic-eye-response. The photocell shall be color-corrected for photopic-eye-response.

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The photocell shall be calibrated with a tungsten lamp of 2,000 watts operated at a color temperature of 2870° K, plus or minus 50° K. An output of less than 58,000 lumens or damage to the source shall constitute failure of this test.

4.6.6 Infrared output. The source shall be supported in a vertical position and operated. A phototube having an S-1 photoemissive response shall be pointed at the source from a distance dictated by the assurance that the phototube is not operated at maximum output capacity. A Corning polished filter number, ETA 2540, color specification number, C8794, 2.6 mm thick or equal shall be placed in front of the phototube, and the output shall be measured. An infrared output of less than 10 percent of the total lumen light output shall constitute failure of this test.

4.6.7 Mechanical shock. The source shall be installed in an approved test fixture mounted in a vertical position, and shall be subjected to 24 shocks of 15 g's plus or minus 10 percent, having a time duration of 6 milliseconds, plus or minus 1.0 millisecond. The maximum g value shall be reached in 3 milliseconds, plus or minus 0.5 millisecond. Twelve shocks shall be applied parallel to the axis of the electrodes and twelve shocks perpendicular to the axis of the electrodes. The source shall be subjected to 3 shock cycles in each plane as follows:

a. The source shall be operated for 2 minutes.

b. After the first minute and while the source is operating, the source shall be subjected to one shock (parallel/perpendicular) to the axis of the electrodes.

c. At the end of the 2 minutes, the source shall be turned "OFF".

d. Three shocks shall be applied within 5 minutes after source is turned "OFF". Any evidence of damage shall constitute failure of this test.

4.6.8 Vibration. The source shall be installed in a test fixture mounted on a vibration table. Precautions shall be taken to prevent magnetic fields from influencing stability of the arc. The source shall be operated during the entire vibration period as follows:

a. Vibrate the source in a direction parallel to the axis of the electrodes with the frequency varying between 15 to 120 Hz at 2.5 g's, plus or minus 0.5 g. The frequency shall change logarithmically from 15 to 120 Hz and return to 15 Hz within 3.5 minutes. Vibration shall continue for not less than 90 minutes; and during this time, the resonant frequency of the anode shall be determined.

b. Vibrate the source at the resonant frequency of the anode 15 minutes. If a resonant frequency cannot be determined the source shall be vibrated at 30 Hz, plus or minus 2 Hz for 15 minutes.

c. Repeat (a) and (b) with the direction of vibration perpendicular to the axis of the electrodes, with duration of (b) being 30 minutes. Any evidence of damage shall constitute failure of this test.

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4.6.9 Altitude. The source shall be placed in the test chamber. The air pressure in the test chamber shall be reduced to the value corresponding to 8,000 feet above sea level and ambient air temperature maintained at plus 35°C. The source shall be ignited and operated continuously for 1 hour at rated power. Failure of the source to ignite and maintain ignition within 3 seconds and operate for 1 hour at rated power, or evidence of damage, shall constitute failure of this test.

4.6.10 Altitude transportation. The source shall be placed in a test chamber and the air pressure in the chamber shall be reduced to the value corresponding to 40,000 feet above sea level with corresponding temperature for this altitude. The chamber shall be maintained at this pressure and temperature for 4 hours. At the end of 4 hours the chamber air pressure shall be returned to the value corresponding to sea level. The source shall then be examined for damage. Any evidence of damage shall constitute failure of this test.

4.6.11 Storage temperature.

4.6.11.1 High temperature storage. The source shall be stored in a test chamber at an ambient temperature of plus 68°C with humidity controlled at 90 to 98 percent for 6 hours. At the conclusion of the storage period the source shall be examined for defects. Any evidence of damage shall constitute failure of this test.

4.6.11.2 Low temperature storage. The source shall be placed in a test chamber and exposed to an ambient temperature of minus 62°C and maintained for a period of 6 hours. At the conclusion of the exposure period, the source shall be examined. Any evidence of damage shall constitute failure of this test.

4.6.12 Operating temperature.

4.6.12.1 High temperature operation. The source shall be mounted in a test fixture. The source and fixture shall be placed in a test chamber and subjected to 2 twelve hour cycles of exposure with relative humidity uncontrolled. Each cycle shall consist of the following:

a. Two hours at plus 52°C.
b. Decrease temperature within one hour to plus 23°C.
c. Maintain plus 23°C for one hour.
d. Increase temperatures within one hour to plus 52°C.
e. Maintain plus 52°C for two hours.
f. Decrease temperature to plus 23°C within one hour.
g. Maintain plus 23°C for one hour.
h. Increase temperature within one hour to plus 52°C.
i. Maintain plus 52°C for two hours.

Immediately upon completion of the 2 cycles the source shall be operated at plus 52°C as specified in 4.6.2. Failure of the source to operate as specified shall constitute failure of this test.

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4.6.12.2 Low temperature operation. The source, installed in a test fixture mounted on a vibration table, shall be placed in a test chamber and cold soaked at minus 54°C for 2 hours and then shall be vibrated at this temperature at its resonant frequencies for 5 minutes with the direction of vibration being perpendicular to the axis of the electrodes. If a resonant frequency cannot be determined the source shall be vibrated at 30 Hz, plus or minus 2 Hz for 5 minutes. The source shall then be ignited while vibrating at 30 Hz and shall be operated for 3 minutes. The source shall then be turned "OFF" and allowed to cool for 12 minutes in the test fixture. The above cycle, not including the 2-hour cold soak, shall be repeated in its entirety until the source has been vibrated a total of 80 minutes. Any evidence of damage shall constitute failure of this test.

4.6.13 Quartz bulb. The optical quality of the quartz bulb shall be tested on a shadowgraph or optical transit as follows (the source shall not be operated during the test) and the test data recorded:

a. A fixture shall be used for mounting the source with the electrodes in a horizontal attitude. The fixture shall permit rotation about the centerline of the electrodes. The fixture shall also be capable of being rotated in azimuth about a point, formed by the apex of the conical surface of the cathode. The source shall be positioned in the fixture so the apex of the conical surface of the cathode is in the same vertical centerline as the fixture rotates about in azimuth and so the optical line of sight through the shadowgraph or optical transit is perpendicular to and passes through the longitudinal axis of the source.

b. Rotate the source 360° about its longitudinal centerline without changing the azimuth position.

c. Rotate the source 30° in a clockwise direction around the vertical centerline through the apex of the conical surface of the cathode and perpendicular to the longitudinal axis and repeat step (b) above.

d. Rotate the source in a counterclockwise direction 22 1/2° around the same centerline as in step (c) and repeat step (b) above.

e. Repeat step (d) in 22 1/2° increments for a total of 67 1/2°. Any apparent shift of the cathode more than 1.0 millimeter radius shall constitute failure of this test.

4.6.14 Electromagnetic compatibility. The source, designed for the reduction of electromagnetic interference in accordance with 3.5.14, shall be tested in accordance with MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.6.15 Life. The source, mounted in a test fixture, shall be ignited with an input of 16 volts dc for the first 150 hours and 22 volts dc for the second 150 hours. After ignition the input voltage to the source shall be maintained at 22 volts dc and the tests conducted as follows, except when performing step (d) below:

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a. Perform (a) of 4.6.8, except that the direction of vibration shall be perpendicular to the axis of the electrodes and the duration shall be 10 minutes with no determination of resonant frequencies.

b. Stop vibration and continue to operate the source for 10 minutes.

c. Repeat (a) and (b) above until 60 minutes of source operation has been accumulated.

d. Perform 4.6.3.

e. Turn the source "OFF" for 10 minutes, with the airflow continued.

f. Repeat (a) through (e) above until 20 hours of source operation have been accumulated.

g. Ignite and operate the source for 50 minutes after which (d) and (e) shall be performed.

h. Repeat step (g) above until 300 hours of source operation have been accumulated. The 300 hours shall include source operation from previous tests.

i. Perform 4.6.5.

Failure of a source to ignite as specified in 3.5.1, source life of less than 300 hours, lumen output of less than 70 percent of its initial value after 300 hours of operation, or damage to the source shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each source shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each source shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.4.1 Technical literature. Technical literature shall be packaged method 1, C-1 of MIL-P-116.

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5.1.1.4.2 Pre-packaging. Each source shall be individually packaged Method III as follows: Assemble the components of the special designed container around the source in accordance with SC-DL-613912. The container shall be taped around the seams with tape conforming to PPP-T-76.

5.1.1.4.3 Source. Each source pre-packaged as specified in 5.1.1.4.2 shall be cushioned within class 2 polyurethane, performed, conforming to MIL-P-26514, not less than 2 inch thickness on all surfaces. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c weather-resistant. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. The source shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited transit storage.

5.2 Packing. Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting super-imposed loads during shipment and storage without damaging the containers or their contents.

5.2.1 Level A. A quantity of sources, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. A quantity of sources, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621 style 4, class 1; PPP-B-585, style 2 or 3c, class 3; PPP-B-640, class 2, style E; or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure and strapping or reinforcing shall be in accordance with the applicable container specification or appendix thereto except that fiberboard shipping containers shall be reinforced by pressure-sensitive filament-tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. A quantity of sources, packaged as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 1, style E, or PPP-B-636, type CF, class domestic. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing of boxes conforming to PPP-B-640 shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the box specification.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The source is intended for use as a radiation source in a searchlight conforming to MIL-S-52444.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity and schedule for first article testing for all line items (see 4.3).

c. Production delivery schedule (see 4.5.2.3.1).

d. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

e. Technical publication as required.

f. Actions required relative to Group C failure (see 4.5.3).

6.3 Definitions. For the purpose of this specification the following definitions apply.

6.3.1 Damage. Damage is defined as deformation, loosening, breakage, corrosion, or change of fit of any component or part, or any physical condition resulting in noncompliance of the source with the provisions of this specification.

6.3.2 "g". g is defined as an acceleration or deceleration of 32.17 feet per second per second.

CUSTODIANS:

Army-EL
Navy-EC
Air Force-84

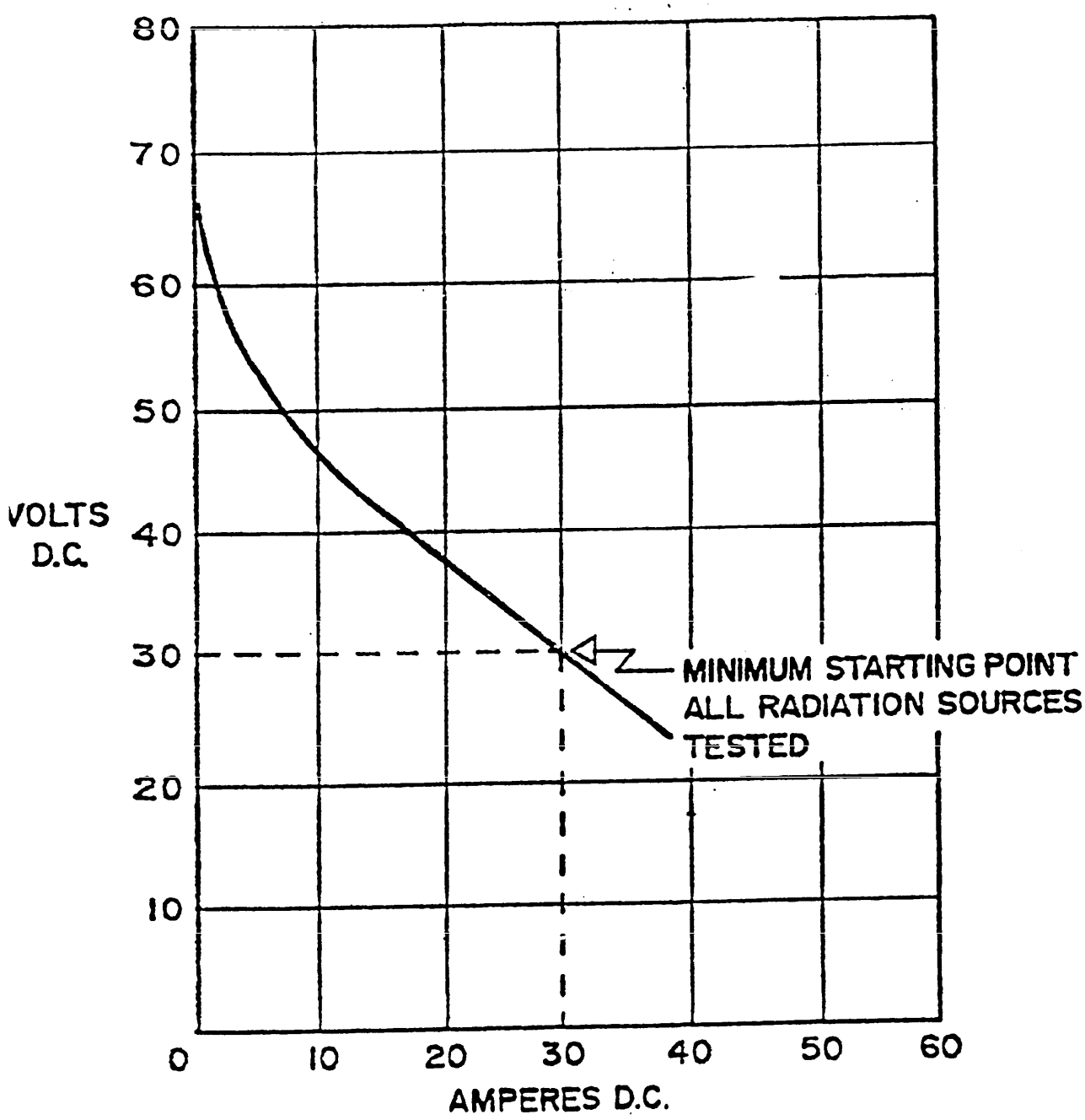
User activity:
Navy - MC

PREPARING ACTIVITY:

Army-EL

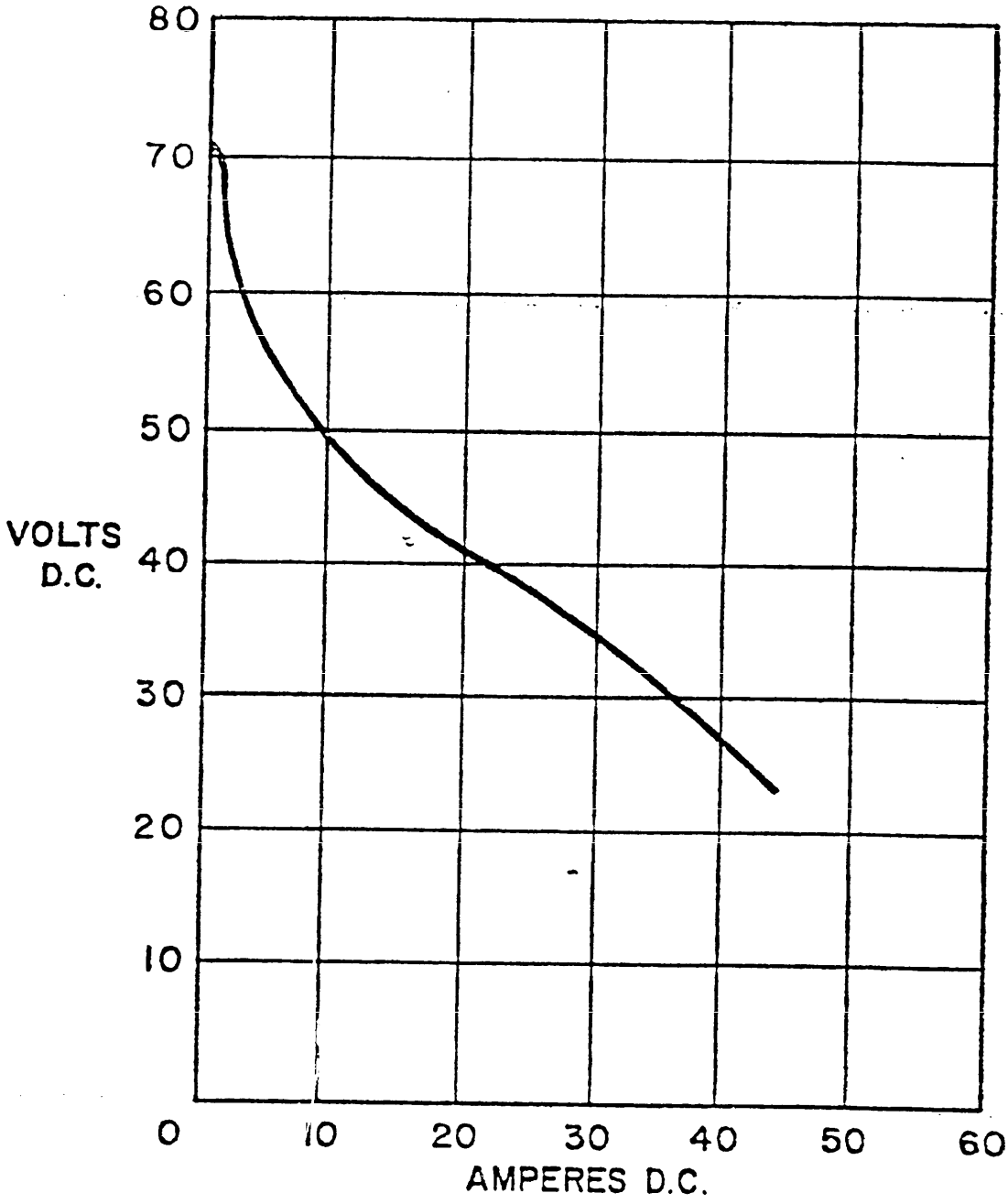
Project Number: 5855-0010

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D.C. VOLT-AMPERE CHARACTERISTIC
CONVERTER-BOOSTER STARTER
(FIG 1)

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D.C. VOLT-AMPERE CHARACTERISTIC
BOOSTER-STARTER 24 VOLT D.C. INPUT
(FIG 2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Fort Monmouth, New Jersey 07703

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